

Techno Terminal - System Capabilities & Features Contract

This document provides a formal overview of the core capabilities, modules, and user workflows supported by the Techno Terminal STEM Education Center Management Platform.

1. Core Modules

1.1. CRM (Customer Relationship Management)

The CRM module acts as the central directory for students, parents, and staff, offering granular profile tracking and advanced search capabilities.

- **Workflow Highlights:**
 - Detailed student profile management with contact information, emergency details, parent linkages, and enrollment status.
 - Parent contact directories linked directly to students for billing and notification routing.
 - Advanced search and filtering tools for quickly locating individuals by name, contact details, status, or tag.
 - Waiting list tracking and status lifecycle management for onboarding prospective students.

1.2. Academics & Course Administration

This module coordinates the center's educational offerings, standardizing course structures, scheduling, and learning progression tracks.

- **Workflow Highlights:**
 - Course curriculum creation with support for different academic tiers, levels, and prerequisites.
 - Group scheduling and directory listing, allowing students to be grouped by level, day, time, or assigned instructor.
 - Enriched class session planning, mapping out future academic years with defined time slots and capacities.

1.3. Student Enrollments

Facilitates onboarding, migrating, and offboarding students across different courses, tiers, and class groups with automatic conflict resolution.

- **Workflow Highlights:**
 - Seamless group-to-group migrations with automated conflict checking for schedules and pricing tiers.
 - Enforcing seat limits and waitlist triggers to avoid overbooking class sessions.
 - Historical auditing of enrollment transitions and status logs.

1.4. Attendance Tracking

Streamlines the tracking and verification of student attendance across all class sessions with automated alerts.

- **Workflow Highlights:**
 - Easy-to-use digital attendance sheets for instructors to record attendance, absences, and make notes.
 - Automatic dashboard widgets summarizing attendance rates per group or per student.
 - Linking attendance records directly to notification triggers (e.g., alert parents on absent status).

1.5. Finance & Billing

Provides robust financial accounting, tracking income, invoice generation, receipts search, and debtor management.

- **Workflow Highlights:**
 - Dynamic invoice creation, discount processing, and payment recording.
 - Receipt generation, ledger entries, and historical receipt search capabilities matching backend database records.
 - Automated debtor tracking, highlighting unpaid student balances and identifying the top debtors.
 - Session-based instructor payout calculations and class-hour verification.

1.6. Competitions & Team Management

Coordinates external and internal competitions, tracking team formation, registrations, and student accomplishments.

- **Workflow Highlights:**

- Competition event creation, registration management, and scheduling.
- Team assignment workflows, mapping students and coaches to designated teams.
- Historical records of competition performance, medals, and achievement tracking.

1.7. HR & Staff Management

Manages staff accounts, roles, access levels, and active work assignments across the center.

- **Workflow Highlights:**

- Directory of all employees, including instructors, administrative personnel, and system admins.
- Secure credential management and profile activation/deactivation workflows.
- Workload assignment and teaching schedule integration.

1.8. Analytics & BI Reports

Generates business intelligence, highlighting financial health, student retention, growth trends, and operation metrics.

- **Workflow Highlights:**

- Daily business dashboards showing active student counts, today's payments, and upcoming schedules.
- Detailed analytical reports summarizing monthly and weekly growth.
- Interactive charts visualizing financial collections and performance indicators.

1.9. Notification Engine

Handles automated messaging, system alerts, and reports via SMTP Email and WhatsApp integration.

- **Workflow Highlights:**

- Configurable notification templates with variable placeholders (e.g., student name, balance due).
- Background dispatch queue for parent alerts, enrollment confirmations, and payment receipts.
- Daily, weekly, and monthly business reports containing session attendance per instructor, list of payments, unpaid students, and top debtors.

1.10. Authentication & Security

Ensures secure platform access, role separation, and session management using advanced authentication flows.

- **Workflow Highlights:**
 - Multi-Factor Authentication (MFA) enrollment and status tracking on user profiles.
 - Role-based route protection separating administrators, system administrators, and instructors.
 - Security audit triggers, including real-time administrative login notifications.

2. Role-Based Access Control (RBAC)

The system enforces strict permission levels based on user roles:

- **System Administrators:** Full system configuration, database access, settings customization, user invitations, and security log reviews.
- **Administrators:** Complete access to business modules (CRM, Finance, HR, Enrollments, Analytics, Notifications) to manage daily operations.
- **Instructors:** Read/write access limited to Academics, Attendance, Teams, and Competitions. Blocked from administrative settings, finance pages, and overall reports.